TSANG Cheuk Nam

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Summary

Final-year HKUST Industrial Engineering & Engineering Management student (Extended Major in AI). Won the "Reinventing Insurance for an AI-native World" track (200+ participants) with an original insurance methodology, then built the complete system. Designs AI systems and the processes around them, directing AI coding agents (Claude Code, Codex) that write about 95% of the code. Coursework complete; available full-time now.

Education

The Hong Kong University of Science and Technology (HKUST) Hong Kong
BEng in Industrial Engineering & Engineering Management; Extended Major in AI *Sep 2022 – Dec 2026 (expected)*

Honors: Winner, "Reinventing Insurance for an AI-native World" track (sponsored by bolttech), Hong Kong AI Summit: BUILD ACROSS Hackathon, 200+ participants (Oct 2026)

Coursework: Statistical Machine Learning; Generative AI and LLMs for Operations Research; Stochastic Models

Projects

Budget-Based Microinsurance | *Hackathon-winning methodology* · Rust, TypeScript, Python Oct 2026

- Designed an original AI-native insurance methodology: customers pick a monthly price and AI builds a personal cover, answering long-standing pain points for insurers, partners and customers.
- Built the payout engine to settle automatic payouts for 1 million policies in 1.6 seconds, against a 5-second target.

LLM Brand Recommendation Benchmark | *AI-native market research* · Python, SQLite, LLM APIs Sep 2026

- Designed an AI-run benchmark of which brands AI assistants recommend, setting every test category at the level customers shop for, so results compare cleanly and new categories fit in without rework.
- Measured that an AI assistant's top-ten brand recommendations matched verified brands 92.5% of the time across 51 product categories.

Using LLMs for Optimization | *Final Year Project, Experiment & Benchmarking Lead* Completed 2026

- Led the experiments behind a reliable way for LLMs to solve business routing problems: the LLM plans and a standard optimizer does the exact work, matching an LLM that writes the whole program at a fraction of the cost and nearly twice as fast.
- Showed the planner approach found a valid routing solution in every trial, on average within 1% of a dedicated solver's best, and matched the solver on public benchmark tests.

PairUp | *Peer-to-peer marketplace that AI agents run* · TypeScript, Rust, AWS Oct 2026

- Designed it so any compatible AI agent can act for its user through a Model Context Protocol (MCP) server, live on AWS with Amazon Bedrock.
- Decided agents should not haggle: an open-source pricing rule settles every deal within both sides' private limits, so no model can be talked into revealing a limit and no deal costs extra model calls.

AI Agent Collusion Research | *AI agent safety research, Apart Research sprint* · Python, vLLM Oct 2026

- Originated and led a pre-registered, four-LLM study of AI agents that formed a cartel over chat: closing the chat ended it, and no model rerouted the collusion into the routine actions it still controlled.
- Designed the study around lying to agents about which channels are real: three of four models kept using a secretly dead private channel, and in all four, agents claimed covert signalling the logs show never happened.

9UPPER Online | *Multilingual real-time web game* · Next.js, TypeScript, Supabase Mar 2026 – Present

- Built it with AI coding agents and grew it through SEO, GEO and AEO to over 500 visitors a day and about 600 Google clicks a month, ranking in Google's top five on average; ChatGPT and Claude search recommend it.

Skills

LLM engineering: RAG, LoRA fine-tuning, Hugging Face, vLLM serving, prompt engineering, multimodal (vision)

AI agents: MCP servers, harness and context engineering, evals, guardrails, multi-agent evaluation, model routing

AI-native transformation: process redesign, pain-point mapping, human-in-the-loop workflows, AI cost control

Agentic coding: Claude Code, Codex, spec-driven development, multi-agent orchestration, agent skill design

Software: Python, TypeScript, React, Rust, PostgreSQL, REST APIs, AWS (Bedrock), Docker, Linux, Git

Languages: Cantonese (native), English (fluent), Mandarin (fluent)